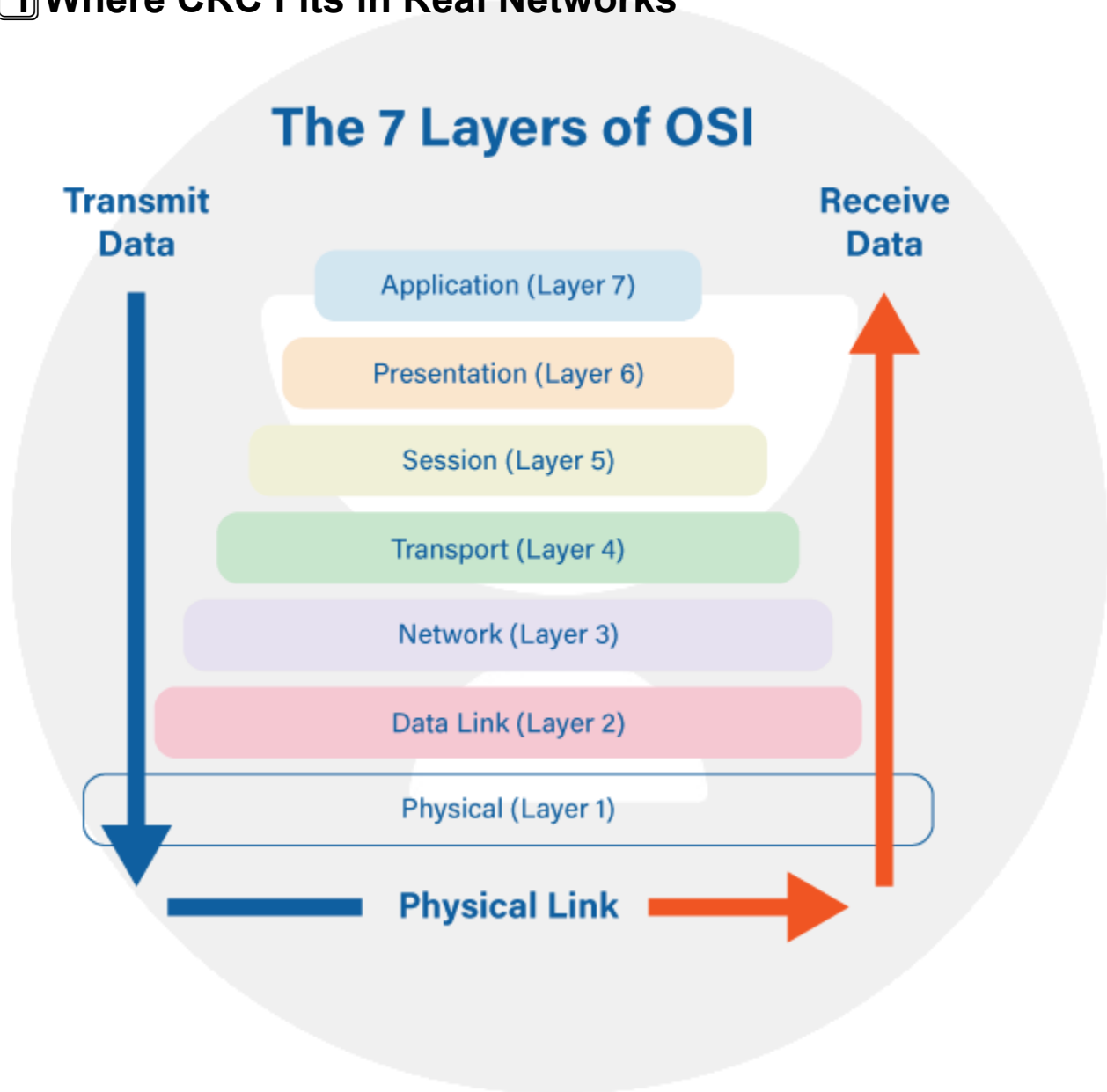


Cyclic Redundancy Check (CRC) and Error Control – Visual + Concept

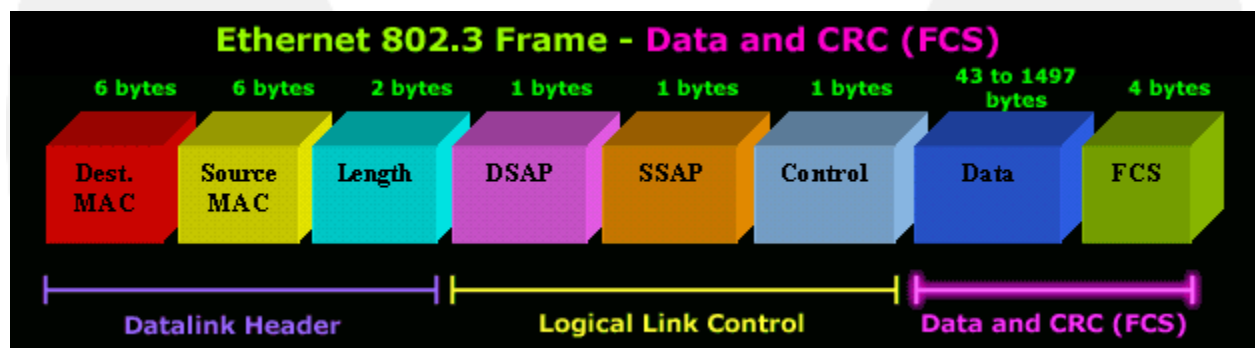
1 Where CRC Fits in Real Networks



Preamble and SFD (8)	MAC header			DATA and PAD (46 to 9216 [#])	FCS
	DA (6)	SA (6)	TYPE/LENGTH (2)		
Ethernet V2 format frame	TYPE= 0x05DD or larger			DATA	(PAD)
	LENGTH= 0x0000 to 0x05DC			DATA	(PAD)
802.3 format frame	LLC header			SNAP header	
	DSAP (1)	SSAP (1)	CONTR OL (1-2)	OUI (3)	PID (2)
Frames in other formats	TYPE= other than the above			DATA	
				DATA	

(n): Field length (units: octets)

#: The maximum length of the DATA and PAD field is 9216 only when the frame format is Ethernet V2. For 802.3 and other formats, the maximum length is 1500.



CRC is primarily implemented at the **Data Link Layer (Layer 2)** of the OSI model.

In real systems:

- Ethernet uses **CRC-32**
- PPP uses CRC
- HDLC uses CRC
- The CRC value is stored in the **FCS (Frame Check Sequence)** field

It is **error detection only**, not correction.

2 Error Detection vs Correction (Quick Mathematical Recap)

Capability	Required Minimum Hamming Distance
Detect D errors	$(d_{\min} \geq D + 1)$
Correct D errors	$(d_{\min} \geq 2D + 1)$

If:

- $(d_{\min} = 2)$ → can detect 1-bit error
- $(d_{\min} = 3)$ → detect 2-bit errors, correct 1-bit error

CRC typically guarantees strong **detection**, but not correction.

3 Binary CRC Procedure (Step-by-Step Logic)

Definitions

Term	Meaning
n	Dataword length
k	Generator length
k-1	Number of appended zeros
R	CRC (remainder)
C	Codeword

Sender Side Algorithm

1. Choose generator G (length k).

2. Append $k-1$ zeros to dataword.
3. Perform modulo-2 division (XOR division).
4. Remainder ($k-1$ bits) = CRC.
5. Replace appended zeros with CRC.
6. Transmit codeword.

Receiver Side Algorithm

1. Receive full codeword.
2. Divide by same generator.
3. If remainder = 0 → Accept.
4. If remainder $\neq$ 0 → Reject.

The receiver remainder is called the **Syndrome**.

4 Modulo-2 Division Mechanics

In CRC:

- No carries
- Subtraction = XOR
- $1 \oplus 1 = 0$
- $1 \oplus 0 = 1$
- $0 \oplus 0 = 0$

Division continues only while leading bit is 1.

5 Polynomial Representation (Exam Shortcut 🚀)

Binary strings are treated as polynomials over $GF(2)$.

Example:

Binary:

1011

Polynomial form:

[
 $x^3 + x + 1$
]

Polynomial Steps

1. Let generator be degree r .
2. Multiply message $D(x)$ by (x^r) .
3. Divide by $G(x)$.
4. Remainder = $R(x)$.
5. Codeword:
[
 $C(x) = x^r D(x) + R(x)$
]

6 Why Polynomial Method is Powerful in MCQs

In many GATE-style questions:

- The shifted part $(x^r D(x))$ remains unchanged.
- Only last r bits differ.
- Immediately eliminate options that change leading terms.

Example rule:

If generator degree = 3 $\rightarrow$ CRC must be exactly 3 bits.

If remainder = 1 $\rightarrow$ CRC = 001 (pad to 3 bits).

7 Why CRC is So Powerful (Practical Strength)

CRC can:

- ✓ Detect all single-bit errors
- ✓ Detect all double-bit errors (if generator has ≥ 3 terms)
- ✓ Detect all odd number of bit errors (if generator contains factor $x+1$)
- ✓ Detect burst errors shorter than generator length

But it **cannot guarantee detection** if:

Corrupted message is still divisible by $G(x)$.

8 Comparison With Other Techniques

Technique	Detect	Correct	Practical Use
Simple Parity	Odd errors	✗	Rare
2D Parity	1-bit correct	Limited	Memory systems
Checksum	Moderate	✗	Transport layer
CRC	Very strong	✗	Data link layer
Hamming Code	Limited	1-bit	ECC memory

CRC dominates networking because:

- Burst errors are common
- CRC handles burst errors extremely well
- Overhead is reasonable
- Implementation is efficient in hardware

9 Understanding Burst Errors Visually

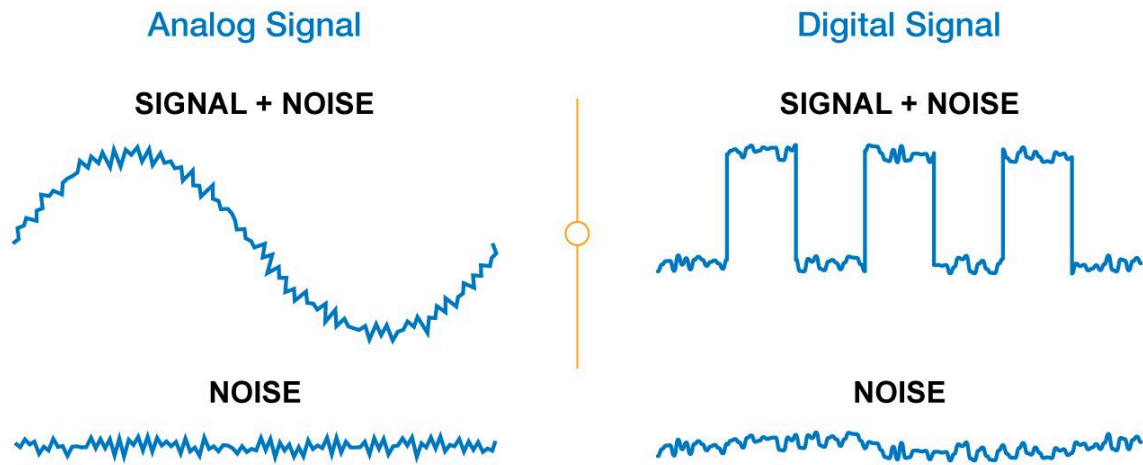


FIGURE 1. Noise in Analog and Digital Signals

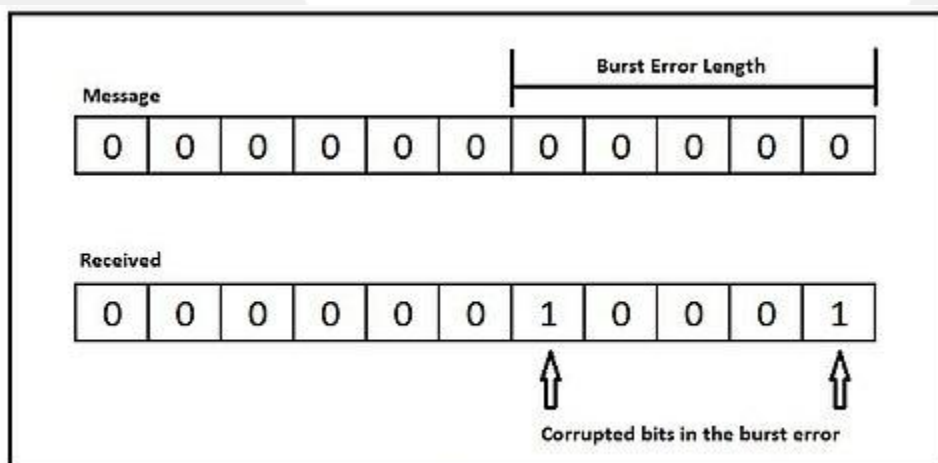
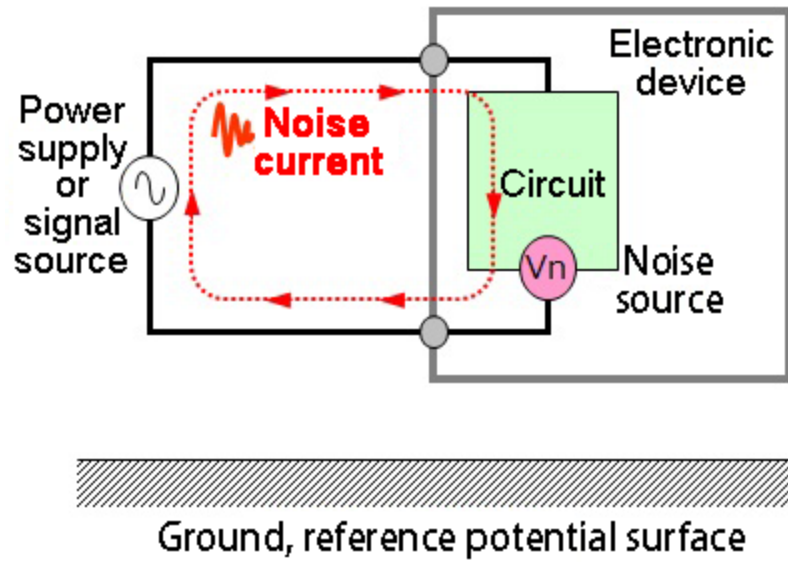


Figure 1



Modern high-speed links experience noise lasting multiple bit durations.
Thus burst errors are far more common than single-bit flips.
CRC is designed specifically to detect these.